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To cite this article: Xin Zhao, Jin Ya Liu & Xin Ming Guo (2025) Multidimensional characteristics of design thinking and project-based learning (DT-PBL) in higher education: a grounded theory study, European Journal of Engineering Education, 50:6, 1285-1304, DOI: [10.1080/03043797.2025.2584060](https://doi.org/10.1080/03043797.2025.2584060)

To link to this article: <https://doi.org/10.1080/03043797.2025.2584060>



Published online: 20 Nov 2025.



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Multidimensional characteristics of design thinking and project-based learning (DT-PBL) in higher education: a grounded theory study

Xin Zhao^a, Jin Ya Liu^b and Xin Ming Guo^{a,b}

^aSchool of Arts, Anhui University of Finance and Economics, Bengbu, People's Republic of China; ^bCentre for Instructional Technology and Multimedia, Universiti Sains Malaysia, Penang, Malaysia

ABSTRACT

This study investigates the underlying mechanisms of Design Thinking and Project-Based Learning (DT-PBL) in engineering education through a grounded theory approach. Based on classroom observations and semi-structured interviews, four interdependent dimensions—self-confidence, reflective interaction, knowledge integration, and iterative refinement—were identified, forming a closed-loop learning ecology. The findings reveal that DT-PBL operates as a recursive, dynamic process: self-confidence drives participation, reflection deepens understanding, and iteration progressively transforms knowledge into actionable insights. Beyond generating project deliverables, DT-PBL cultivates reflective judgment, integrative thinking, and sustainable problem-solving competencies vital for contemporary engineering practice. This study contributes a theoretically grounded and pedagogically actionable framework for advancing DT-PBL implementation in higher education.

ARTICLE HISTORY

Received 13 February 2025
Accepted 29 October 2025

KEYWORDS

Design thinking and project-based learning; social innovation education; grounded theory; higher education

1. Introduction

In the rapidly evolving knowledge-based economy, societal progress increasingly depends on educational systems that move beyond traditional modes of knowledge transmission, placing greater emphasis on fostering innovation, adaptability, and interdisciplinary collaboration. Conventional engineering education has historically prioritised cognitive and technical competencies, or 'hard skills,' such as mathematical reasoning and domain-specific expertise. While these skills remain essential, such approaches frequently underemphasise creativity, critical thinking, and teamwork – 'soft skills' that are critical for addressing complex real-world problems and adapting to rapidly changing professional environments.

To address these limitations, engineering education must transition toward more integrative and experiential learning models. One widely adopted approach is Project-Based Learning (PBL), which emphasises student-led, collaborative problem-solving within authentic, real-world contexts. PBL is recognised as a powerful pedagogical strategy that bridges classroom learning with societal needs, while fostering transferable skills, such as communication, initiative, and reflective thinking (Kokotsaki, Menzies, and Wiggins 2016; McDonnell-Naughton and Păunescu 2022). When explicitly aligned with innovation-oriented learning objectives, PBL can serve as a lever for targeted educational reform, enhancing the societal relevance of engineering curricula (Maldonado-Mariscal and Schröder 2023).

Complementing PBL, Design Thinking (DT) has emerged in education as a transformative, human-centered, and iterative framework that emphasises empathy, ideation, prototyping, and testing as

the core stages to address issues such as inflexible operations and poor learning experiences in the PBL phase (Holst 2023). As a pedagogical model, DT positions students as co-creators of knowledge rather than passive recipients, fostering active learning and bridging the gap between academic curricula and the evolving demands of the professional world (Caton et al. 2022). While PBL focuses on inquiry and project execution, DT introduces a structured creative process that encourages problem reframing, in-depth user research, and rapid iteration, providing an additional layer of creative exploration and user empathy (Manzini 2015; Wei, Zhang, and Lin 2023).

Recent scholarship has explored integrating DT and PBL into a hybrid DT-PBL model, designed to combine DT's emphasis on creativity and user-centeredness with PBL's experiential, collaborative framework (Lebid and Shevchenko 2020). This hybrid model has shown promise in enhancing students' cognitive engagement, communication skills, motivation, and self-efficacy (Chandrasekaran et al. 2013; Jia, Jalaludin, and Rasul 2023). In engineering education, the DT-PBL model is gaining traction as traditional lecture-based approaches often struggle to prepare students for the ambiguity, interdisciplinarity, and iterative problem-solving required in real-world design and innovation contexts (Collins and Chiaramonte 2017). Engineering students, in particular, benefit from DT-PBL's structured approach to navigating ill-defined problems, fostering effective teamwork, and developing both technical competencies and reflective capacities – skills that are frequently underdeveloped in conventional curricula (Wei, Zhang, and Lin 2023).

Additionally, the advent of digital technologies has enabled DT-PBL to evolve into more flexible, personalised, and interactive formats. Digital platforms facilitate distributed collaboration, real-time feedback, and iterative prototyping, thereby enriching the pedagogical process and enabling deeper engagement with human and technological systems (Caton et al. 2022). Thus, DT-PBL can be understood as a strategic innovation framework capable of addressing diverse educational challenges in the digital era (Collins and Chiaramonte 2017; Zhao et al. 2023; Henriksen, Mehta, and Mehta 2019).

Despite these encouraging developments, significant gaps remain. The theoretical foundations of DT-PBL are not yet fully articulated, and practical challenges – such as outcome uncertainty, insufficient teacher preparation, and the complexity of assessing open-ended projects – remain inadequately addressed (Zhao et al. 2023). Moreover, there is a lack of empirical research from the student perspective, particularly in engineering education. Conventional assessment methods often fail to capture the full spectrum of learning outcomes in DT-PBL, making it difficult to evaluate 'meaningful and measurable change.' In this study, the term specifically refers to the quantifiable enhancement of students' soft skills, critical reflection, and ability to address real-world problems.

This study aims to explore the defining characteristics of DT-PBL and their impact on students' learning processes, outcomes, and skill development. Specifically, the objectives are threefold: (1) Construct a theoretically grounded model of DT-PBL by systematically identifying its multidimensional characteristics and analysing their impact on learning outcomes, attitudes, and developmental factors; (2) Investigate students' lived experiences and perceptions during DT-PBL by examining how they engage in reflective interaction, integrate knowledge, and refine solutions through iterative cycles. (3) Develop actionable guidelines for implementing effective DT-PBL pedagogies that enhance student engagement, foster innovation, and improve practical learning outcomes.

Building on these objectives, the study addresses the following research questions:

RQ1 (Model Development): What characteristics and mechanisms of DT-PBL can be empirically identified through grounded theory analysis, and how do these form a coherent pedagogical model in engineering education?

RQ2 (Learning Dynamics): How do students' learning processes unfold in DT-PBL, particularly in relation to reflective interaction, knowledge integration, and iterative refinement across different project phases?

RQ3 (Experiences and Outcomes): What kinds of experiences and outcomes do students achieve through DT-PBL, and how can these be understood as meaningful and measurable change in terms of reflective capacity and real-world problem-solving ability?

2. Literature review

To establish a solid foundation, this study conducted a systematic literature review across Scopus, Web of Science, ERIC, and Google Scholar, using the search terms ‘Design Thinking,’ ‘Project-Based Learning,’ ‘DT-PBL,’ and ‘Higher Education.’ The review covered publications from 2015 to 2025, with earlier foundational works included for theoretical context. Of 156 retrieved records, 61 were retained after relevance screening and quality appraisal. This ensured both breadth and selectivity, avoiding the tendency – common in emerging fields – to cite every mention of DT-PBL without critical synthesis.

DT emerged from professional design practice, notably through IDEO and Stanford’s d.school (Johansson-Sköldberg, Woodilla, and Çetinkaya 2013). In education, it is defined as a human-centered, iterative process encompassing empathy, problem reframing, ideation, prototyping, and testing (Buehring and Liedtka 2018; Henriksen, Mehta, and Mehta 2019). Grounded in self-efficacy and innovation education theories, DT emphasises learning through cycles of experimentation and feedback that build mastery and confidence in navigating uncertainty (Caton et al. 2022; Groeger et al. 2019). Simultaneously, innovation education theory interprets DT as a catalyst for reframing and divergent thinking, positioning learners as co-creators who cultivate creative confidence and adaptive expertise (Grau and Rockett 2022).

However, the educational adoption of DT remains uneven. Scholars warn that it is often reduced to procedural toolkits emphasising templates over cognitive transformation (Johansson-Sköldberg, Woodilla, and Çetinkaya 2013). Without a robust theoretical grounding, DT risks producing short-term engagement without enduring conceptual change (Razzouk and Shute 2012). This gap underscores the need to articulate how DT’s pedagogical potential translates into sustainable learning mechanisms.

By contrast, PBL has a long-established tradition in higher education, particularly in engineering disciplines (Kolmos 2009; Rohmaniyah and Asih 2024). PBL centres on authentic engagement with complex problems, collaboration, and tangible project outcomes that integrate disciplinary knowledge with applied problem-solving (Kokotsaki, Menzies, and Wiggins 2016). Its theoretical grounding lies primarily in social learning theory and systems thinking, which together explain how learning emerges through collaboration within dynamic socio-technical contexts. Social learning theory highlights peer interaction, modelling, and shared reflection as mechanisms for transforming collective participation into individual competence (Bandura 2001). Systems thinking complements this view by framing PBL as engagement with interdependent and evolving systems, prompting learners to analyse feedback loops, emergent behaviours, and ethical implications (Kokotsaki, Menzies, and Wiggins 2016).

While PBL strengthens critical thinking and teamwork, its procedural rigidity can limit creativity and reflective inquiry. Emphasis on deliverables and outcomes often constrains divergent exploration and user-centered thinking (Sisamud, Chatwattana, and Piriya-surawong 2023). From the lens of innovation education, such a structure may inhibit the development of adaptive expertise and creative confidence – key competencies for transformative learning (Kokotsaki, Menzies, and Wiggins 2016). Moreover, systems thinking reveals that traditional PBL tends to fragment problem-solving into discrete tasks, overlooking the socio-technical interdependencies central to authentic innovation (Jia, Jalaludin, and Rasul 2023).

Accordingly, DT and PBL should be viewed as complementary pedagogical frameworks. PBL offers structure, authenticity, and contextual engagement, whereas DT contributes empathy, iteration, and creative reframing (Indrianti and Kurniawan 2022; Jia, Jalaludin, and Rasul 2023). Conceptually, their integration – DT-PBL – can be understood as a multi-layered learning system underpinned by four interconnected theories: self-efficacy, social learning, innovation education, and systems thinking. Iterative feedback loops strengthen agency and confidence (Bandura 2001; Grau and Rockett 2022); collaborative reflection fosters shared cognition; experimentation and

reframing nurture creative adaptability (Collins and Chiaramonte 2017); and systemic reasoning links empathy to socio-technical awareness (Jia, Jalaludin, and Rasul 2023).

Emerging empirical evidence supports this synthesis. Studies report gains in creativity (De la Fuente, Carbonell, and LaPorte 2019), collaborative problem-solving (Rohmaniyah and Asih 2024), and motivation (Jia, Jalaludin, and Rasul 2023). Research linking DT-PBL to self-efficacy and innovation frameworks demonstrates enhanced persistence and creative agency (Grau and Rockett 2022), while social and systems-oriented studies highlight improved shared cognition and socio-technical understanding (Indrianti and Kurniawan 2022). Nonetheless, most evidence remains small-scale and descriptive, offering limited insight into the mechanisms and boundary conditions that determine DT-PBL's effectiveness.

In engineering education, these theoretical intersections are particularly salient. Traditional curricula have long been criticised for privileging analytical rigour over creativity, empathy, and socio-technical awareness (Martin 2020). However, contemporary engineers increasingly confront 'wicked problems' such as sustainability and human-machine interaction, which demand integrative and reflective thinking. DT-PBL responds to this need by embedding user-centered inquiry within authentic project structures, connecting technical rigour with human and societal relevance (Jia, Jalaludin, and Rasul 2023; Seng and Tan 2024). Recent innovations extend this approach through computational thinking for systems reasoning (Chang et al. 2023), immersive technologies for distributed collaboration (Rohmaniyah and Asih 2024), and capstone projects linking academia with industry and communities (Satrústegui, Gurruchaga, and Irastorza 2021). Despite these promising developments, most implementations remain conceptual prototypes rather than rigorously evaluated pedagogical systems.

In summary, the literature positions DT-PBL as a promising yet under-theorized innovation. It addresses recognised gaps in engineering education by integrating technical proficiency with human-centered and systemic awareness. However, three critical gaps persist: (1) limited theoretical clarity, as DT-PBL is often described but rarely conceptualised rigorously; (2) inconsistent methodological rigour, with many studies relying on small-scale or anecdotal data; and (3) insufficient evidence of long-term impacts, including employability, socio-technical awareness, and sustainable innovation. This study addresses these gaps through a grounded theory approach, conceptualising the multidimensional characteristics of DT-PBL and tracing how students engage in reflective interaction, knowledge integration, and iterative solution refinement. By doing so, it advances from descriptive accounts toward explanatory depth, offering an empirically grounded and pedagogically actionable framework.

3. Methodology and research design

This study employs a grounded theory (GT) approach, a qualitative methodology suited for investigating complex and under-theorized educational phenomena. GT's inductive logic enables researchers to systematically derive theoretical constructs from empirical data, avoiding premature reliance on pre-existing frameworks (Bryant and Charmaz 2007). The iterative cycles of data collection and analysis allow researchers to refine categories until theoretical saturation is reached. This methodological flexibility makes GT particularly valuable in education, where learning processes are multifaceted and context-dependent.

This study was conducted within a senior-level engineering programme at a comprehensive Chinese university, where DT-PBL has been systematically implemented in interdisciplinary project courses. Each course spans one semester and engages students in authentic social and technical challenges that require both disciplinary expertise and cross-functional collaboration. This context was chosen because it reflects current demands in engineering education to integrate hard skills (e.g. technical problem-solving, analytical reasoning) with soft skills (e.g. collaboration, creativity, reflective judgment). The programme's interdisciplinary laboratories and industry-linked projects provided a unique environment for observing DT-PBL in practice.

Sampling followed theoretical sampling principles, with participants selected iteratively to maximise conceptual variation and explanatory depth. This design served two purposes. First, it captured multiple perspectives, which is consistent with GT's emphasis on systemic and socio-cognitive processes. Second, it aligned with engineering education research, which highlights the interplay of students, instructors, and external stakeholders in shaping outcomes (Kolmos 2009). Accordingly, students provided accounts of their learning experiences, teachers offered insights into pedagogical design, and stakeholders contributed perspectives on societal and professional relevance. Sampling and analysis proceeded concurrently until no new conceptual insights emerged, signalling theoretical saturation.

In addition to interviews, classroom observations were integrated into the design. Observations enabled the research team to capture real-time, interactive data, including group dynamics, the use of DT tools, and the negotiation of roles that might not be revealed in self-reported interviews. These observations also provided methodological triangulation, enabling comparisons between participants' interview responses and their actual practices, thereby enhancing the validity of the findings.

To further enhance validity, a multidisciplinary teaching and research team was assembled, including DT specialists, education innovation scholars, and disciplinary experts. Previous research on interdisciplinary pedagogy emphasises the value of epistemic diversity in creating transformative learning environments (Kali 2024). Engaging experts from different fields in coding and interpretation reduced disciplinary bias and increased the credibility of emergent categories (Christensen 2021). Construct validity was strengthened through triangulation across participant groups and team perspectives, while credibility and transparency were supported through reflexive memo-writing and peer debriefing. Together, these strategies ensured that the resulting DT-PBL framework is both empirically grounded and pedagogically robust. The methodological design directly addresses gaps in the literature, specifically the lack of multi-perspective, empirically grounded analyses of DT-PBL in engineering contexts.

3.1. Description of DT-PBL project course context

PBL has been widely recognised as a powerful pedagogical approach for cultivating students' problem-solving capacity by engaging them in authentic, real-world contexts (Kokotsaki, Menzies, and Wiggins 2016). Its structured cycles of inquiry, collaboration, and reflection have been formalised in several authoritative frameworks. For instance, De la Fuente, Carbonell, and LaPorte (2019) and Traifeh et al. (2020) conceptualised PBL as a four-stage cycle: problem identification, concept exploration, concept refinement, and final presentation. Jiang and Pang (2023) extended this into a five-phase model that highlights project awareness, problem definition, solution development, prototyping, and project testing and communication. Sisamud, Chatwattana, and Piriyaawong (2023) similarly proposed the following steps: preparation, topic definition, creation and testing, demonstration, and evaluation. Beyond these, Kolmos (2009) in the field of engineering education underscores the iterative nature of problem framing, ideation, prototyping, and reflection, reinforcing PBL's strength in mimicking the cycles of real-world professional practice. Collectively, these models provide an authoritative basis for selecting PBL as the course's structural backbone (see Table 1).

DT complements PBL by emphasising human-centered, empathy-driven exploration, divergent-convergent ideation, rapid prototyping, and iterative testing (Beckman and Barry 2007). Unlike PBL, which emphasises long-term projects and disciplinary integration, DT explicitly targets the creative and socio-cognitive processes underpinning innovation. Beckman and Barry (2007) highlight the iterative sequence of framing, ideation, prototyping, and reflection as the core of DT pedagogy, while Razzouk and Shute (2012) emphasise DT's role in fostering creativity, confidence, and tolerance for ambiguity. More recent studies (Dell'Era et al. 2020; Verganti 2017) systematize DT strategies into distinct pedagogical practices: Creativity Confidence, Creative Problem Solving, Sprint Execution, and Meaningful Innovation. These practices align with specific phases of project implementation.

Table 1. Review of DT-PBL conceptualizations across the literature.

Phase	De la Fuente, Carbonell, and LaPorte (2019)	Jiang and Pang (2023)	Sisamud, Chatwattana, and Piriyasurawong (2023)	Traifeh et al. (2020)	Dell'Era et al. (2020)/Verganti (2017)	Razzouk and Shute (2012)	Beckman and Barry (2007)	Kolmos (2009)
Exploration	Problem Identification	Project Awareness	Preparation & Topic Definition	Problem Framing	Creativity Confidence	Identify Problem	Observe & Empathise	Problem Analysis
Thinking	Concept Exploration	Solution Development	Creation & Testing	Ideation & Concept Development	Creative Problem Solving	Generate Ideas	Ideate & Conceptualise	Solution Design
Practice	Concept Refinement & Prototyping	Prototyping	Testing	Prototype Development & Iteration	Sprint Execution	Prototype Solutions	Prototype & Iterate	Implementation
Confirmation	Final Concept Presentation	Project Testing & Communication	Demonstration & Evaluation	Final Presentation & Feedback	Meaningful Innovation	Evaluate & Reflect	Test & Reflect	Evaluation & Dissemination

Note: This synthesis identifies commonalities across multiple PBL conceptualizations, revealing a recurring structure of four interconnected phases – Exploration, Thinking, Practice, and Confirmation – while acknowledging nuanced differences in emphasis. This mapping underpins the PBL framework adopted in the present study, which integrates design thinking strategies into each phase to maximise creativity, user-centeredness, and solution viability.

These frameworks are widely cited and empirically validated, providing an authoritative basis for embedding DT practices in higher education (see [Table 2](#)).

The DT-PBL model developed in this study transcends the additive combination of DT and PBL. Rather than layering DT's creativity onto PBL's procedural structure, it fuses their epistemic cores – design as inquiry and project as context – into a unified, reflective system of iterative innovation. Within this framework, PBL's structured phases of Exploration, Thinking, Practice, and Confirmation are dynamically infused with DT's pedagogical strategies of Creativity Confidence, Creative Problem Solving, Sprint Execution, and Meaningful Innovation. This integration establishes a dual-loop mechanism that links creative cognition with engineering execution. The result is a learning system that cultivates both conceptual adaptability and technical precision – enabling students to move seamlessly between creative ideation and rigorous implementation.

Additionally, this integrated framework was implemented in an 18-week interdisciplinary project course designed by a multidisciplinary teaching team comprising two design thinking specialists, two innovative education experts, and two disciplinary instructors. Its iterative implementation over three academic years provided a stable pedagogical environment and a rich, empirical dataset suitable for grounded theory analysis. Situating the research in this context ensures methodological alignment with the study's overarching objectives: uncovering the multidimensional characteristics of DT-PBL, analysing students' lived experiences, and generating a theoretically robust framework that advances scholarship and practice in higher education pedagogy.

3.2. Research sample

This study employed theoretical sampling, a methodological strategy that strategically guides the selection of sampling sites, interview participants, and interview questions to explore emerging categories, understand their interrelationships, and ensure the appropriateness and representativeness of the data (Dunne 2011). Unlike random sampling, theoretical sampling emphasises conceptual relevance over numerical representativeness. Accordingly, participants were selected to capture variation across demographic, academic, and experiential dimensions that could influence perceptions and enactments of DT-PBL. Several factors were considered when constructing the sample: (1) Gender was included to ensure that potential differences in participation, collaboration styles, or perceived challenges were captured, addressing prior research suggesting gendered dynamics in project-based teamwork. (2) Academic major and year of study were recorded to differentiate students' prior exposure to PBL and disciplinary knowledge, which may shape their ability to engage in DT processes. (3) Years of teaching or participation in DT-PBL served as a proxy for experience level, enabling the study to compare novice versus more experienced participants and to trace the learning curve associated with repeated exposure. Stakeholder involvement (e.g. industry or community representatives) was incorporated to capture external perspectives on the societal and professional relevance of DT-PBL projects (see [Table 3](#)).

Table 2. The Design Ideas, Methods, and Objectives of the DT-PBL Model.

Stage	DT Strategy	Methods Applied	Objectives
Exploration	Creativity Confidence	Group discussions, field observations, semi-structured interviews	Encourage creative thinking, build empathy, and foster collective goal-setting by leveraging individual differences.
Thinking	Creative Problem Solving	Brainstorming sessions, case analysis, and guided group dialogue	Generate diverse ideas, develop innovative insights, and define problems through user-centered analysis.
Practice	Sprint Execution	Co-creation workshops, concept boards, prototyping, rapid experiments	Accelerate development, reduce uncertainty, and refine solutions collaboratively with stakeholders to drive effective outcomes.
Confirmation	Meaningful Innovation	Public showcases, stakeholder feedback sessions, and reflective workshops	Redefine significant problems, establish new developmental goals, and ensure the sustainability of solutions.

3.3. Data collection

Multiple data collection methods were employed to enhance both the comprehensiveness and credibility of the findings. Data sources included: (1) systematic classroom observations across the four course phases (teacher–student interactions, peer-group collaboration, and stakeholder engagement); (2) written reflections and feedback posted on the course platform; and (3) semi-structured interviews with three teachers, five students, and two external stakeholders. Each interview lasted 30–60 min, was audio-recorded with the participant’s consent, and was transcribed verbatim. In total, 2,384 min of recordings produced approximately 77,200 words of textual data.

NVivo 14 was used to manage, code, and query the dataset. NVivo was chosen for its recognised ability to support inductive coding, constant comparison, and keyword clustering in grounded theory studies. To ensure methodological rigour, three-quarters of the interview data were randomly selected for initial coding, factor extraction, and theoretical model construction. This strategy balanced efficiency with representativeness, enabling iterative coding cycles to proceed without excessive redundancy. The remaining quarter of the data was withheld to serve as an independent dataset for theoretical saturation testing. According to Aldiabat and Le Navenec (2018), theoretical saturation was confirmed when no substantially new categories or properties emerged from this reserved data, thereby supporting the robustness and completeness of the developed framework.

Internal reliability and validity were further safeguarded through multiple procedures. First, two independent researchers coded a 25% subset of the data in parallel and then compared coding schemes. Inter-rater reliability was calculated using Cohen’s Kappa ($\kappa = 0.82$), which exceeds the commonly accepted threshold of 0.75 for substantial agreement. Any discrepancies were resolved through collaborative discussion, ensuring shared understanding of category definitions. Second, analytic memos were maintained throughout to document coding decisions, category refinements, and the researcher’s reflexivity, thereby strengthening the auditability of the process. Third, triangulation across different data sources (interviews, observations, and written reflections) provided cross-validation, thereby enhancing construct validity.

4. Data analysis and model construction

This study employed qualitative content analysis and keyword clustering techniques. Following grounded theory procedures (Straus 1998), a hierarchical coding approach consisting of three iterative stages was adopted: open coding, axial coding, and selective coding. The constant comparison method and analytic memo writing were employed throughout the process to refine categories, capture theoretical sensitivity, and ensure transparency in decision-making.

4.1. Open coding

In the open coding phase, data were analysed line by line to maximise theoretical sensitivity, rather than relying solely on descriptive coding. Although sentence – and paragraph-level readings were used for contextual interpretation, the primary principle was to adhere closely to the participants’ own words, gradually abstracting them into conceptual labels. Concepts were retained based on their explanatory power and their recurrence across diverse data sources, rather than on frequency alone. For example, notable original statements and emerging codes (those appearing across at least

Table 3. Basic information table of interview samples.

Basic Information	Characteristics	Teacher	Student	Stakeholder	Total
Gender	Male	2	3	1	6
	Female	1	2	1	4
Academic major/year	Sophomore year	–	3	–	3
	Junior year	–	2	–	2

three participants or data types) were consolidated into broader categories. Based on the four learning stages of DT-PBL, 19 initial concepts were identified and collapsed into 12 first-order categories. Table 4 illustrates how the raw data were progressively abstracted into initial concepts, clustered into first-order categories, and ultimately linked to axial themes.

4.2. Axial coding

Following open coding, the disaggregated categories were reassembled by examining causal, contextual, and consequential conditions (Straus 1998). This process allowed the study to move beyond descriptive clustering toward relational explanation. Twelve sub-categories were refined and

Table 4. Examples of DT-PBL open coding.

Phase	Sample Source Data (Illustrative Quotes)	Initial Concept	Category
Exploration	A1: 'After group formation, we noticed significant differences in academic background and personality. Group discussions helped us break the ice.'	c1 Team Differences and Self-Regulation	C1 Collaborative Interaction
Thinking	A2: 'We do not share the same professional level, so misinterpretations happen. We hold seminars and prepare tutorials to share knowledge.'	c2 Team Knowledge Sharing	
Practice	A3: 'Quick feedback in iterations kept refining solutions and strengthened team cohesion.'	c3 Team Collaboration and Social Participation	
Thinking	A4: 'When discussions hit bottlenecks, teachers guided me by asking about my expectations and providing targeted suggestions.'	c4 Teacher Support and Guidance	C2 Teacher–Student Interaction
Practice	A5: 'Community co-creation workshops allowed us to test and adapt solutions directly with users.'	c5 Community Co-Creation and Testing	C3 Social Interaction
Confirmation	A6: 'Project success depended on community members' cooperation and information.'	c6 Community Conditions and Support	
Exploration	A7: 'The Creativity Confidence module helped me integrate quickly and face uncertainties.'	c7 Challenges of Quickly Integrating into the Group	C4 Interpersonal Confidence
Exploration	A8: 'Observation helped us empathise with user difficulties and identify real problems.'	c8 Enhancement of Empathy	C5 Creativity Confidence
Thinking	A9: 'No matter the problem, I kept trying until I solved it.'	c9 Courage to Face Uncertainty and Challenges	C6 Problem-Solving Confidence
Confirmation	A10: 'Positive feedback from experts boosted my confidence.'	c10 feedback enhances confidence in problem-solving	
Exploration	A11: 'Clarifying the project needs early helped us redefine the real problem.'	c11 Clarification of Task and Process	C7 Information Exchange
Thinking	A12: 'Continuous exchange and teacher feedback improved our decision-making.'	c12 Information Sharing and Decision-Making	
Exploration	A13: 'Design methods helped us systematically integrate relevant knowledge.'	c13 Organise and Integrate Knowledge	C8 Knowledge Integration
Thinking	A14: 'Theoretical input from teachers improved real-world application.'	c14 Combined Theoretical and Practical approach	
Practice	A15: 'Iterating with design methods transformed theoretical knowledge into solutions.'	c15 Knowledge Transformation and Utilisation	
Practice	A16: 'Sprint Execution enabled us to iterate on designs quickly.'	c16 Rapid Iteration and Continuous Improvement	C9 Solution Diversity
Exploration	A17: 'Observation and interviews revealed overlooked details.'	c17 Exploration of Problem Scope	C10 Problem Identification & Definition
Practice	A18: 'Testing many solutions – even wrong ones – helped us find better directions.'	c18 Innovative Thinking and Diverse Solutions	C11 Iterative Experimentation
Confirmation	A19: 'Meaningful Innovation helped us set long-term goals and sustain relevance.'	c19 Establishment of mechanisms for project sustainability and long-term impact.	C12 Meaning Update

Note: Codes such as 'A1' refer to anonymized raw data fragments drawn from interviews or observations. The 'c1–c19' tags denote initial coding labels, while 'C1–C12' indicate the emergent axial themes into which first-order categories were clustered. This layered coding system allows transparent traceability from raw data to theoretical abstraction.

elevated into four axial codes: Self-Confidence, Interactive Reflection, Knowledge Integration, and Iterative Process (see [Table 5](#)).

4.3. Selective coding

Selective coding integrated these axial codes into a coherent analytic storyline. The core category identified was ‘DT-PBL as a reflective and iterative pathway for student transformation.’ This construct unifies the analysis by demonstrating that confidence, reflection, and knowledge integration are not separate outcomes, but rather interdependent dynamics within an iterative learning cycle.

To articulate this storyline, the categories were organised within the ‘Awareness → Context → Action’ model (Peng et al. 2022), which resonated strongly with the emergent data. Importantly, this framework was not imposed but emerged through analytic comparison as the representation of the closed-loop learning processes observed in DT-PBL. Specifically, self-confidence provided the intrinsic motivational force for sustained engagement. Interactive reflection and knowledge integration operated as the relational and cognitive mechanisms through which challenges were negotiated and transformed. Iterative processes acted as feedback loops, ensuring that learning was not static but continually redefined in light of practice.

Taken together, the analysis conceptualises DT-PBL as a cyclical, self-reinforcing learning ecology. These dimensions constitute the structural mechanisms of DT-PBL, moving beyond descriptive elements of classroom practice to capture the psychological and processual foundations of the model. The analysis not only provides the empirical basis for a theoretically grounded model (see [Figure 1](#)) but also demonstrates DT-PBL’s unique capacity to cultivate creativity, reflective judgment, and adaptive problem-solving in engineering education, thereby directly addressing RQ1.

Table 5. Axial coding results.

Self-Confidence	C4 Creativity Confidence	Through creativity training and practice, students gain confidence in proposing innovative solutions when facing uncertainty and challenges.
	C5 Interpersonal Confidence	In teamwork and social interactions, students gradually build confidence in communicating and collaborating with others.
	C6 Problem-Solving Confidence	Through solving real-world problems, students develop confidence in their problem-solving abilities, enabling them to tackle complex issues effectively, both independently and in teams.
Interactive Reflection	C1 Team Interaction	Through interaction and collaboration within the team, students share insights and experiences, promoting collective learning and growth.
	C2 Teacher–Student interaction	Teachers guide and provide feedback during the course, helping students engage in deep thinking and reflection, thereby enhancing learning outcomes.
	C3 Social Interaction	By interacting with community members and stakeholders, students gain honest feedback and insights, improving the practicality and effectiveness of their solutions.
Knowledge Integration	C7 Information Exchange	During the learning process, students enhance their comprehensive understanding of issues by collecting, analysing, and exchanging information.
	C8 Knowledge Transformation	Through practice, students combine theoretical knowledge with the need to solve real-world problems to develop systematic solutions and complete the process of knowledge internalisation.
	C9 Practical Test	Students propose a range of potential solutions, and through discussion and evaluation, they select the most optimal solution, testing its feasibility through practical application.
Iterative Process	C10 Problem Identification and Definition	Through in-depth social exploration, students identify and define key problems, providing clear direction for subsequent design and solutions.
	C11 Trial and Error	In the problem-solving process, students learn and improve through continuous trials and errors, finding the best solution.
	C12 Meaning Update	During learning and practice, students continuously reflect on and update their understanding of the problem and the significance of the solutions, ensuring their sustainability and long-term development.

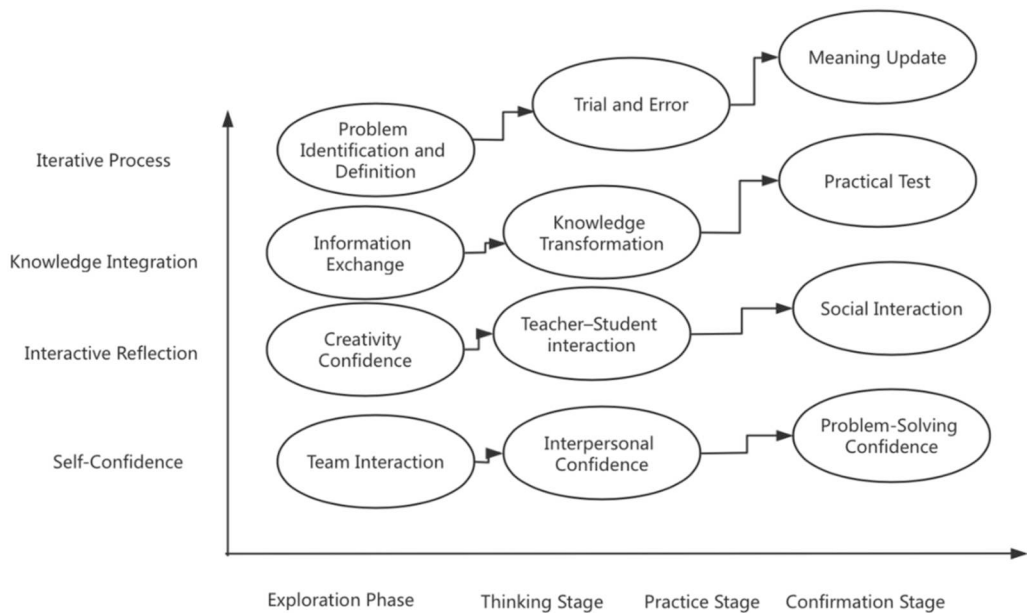


Figure 1. The multidimensional characteristics model of DT-PBL.

5. Results

Building on the coding process, this section elaborates on how the four axial categories – Self-Confidence, Interactive Reflection, Knowledge Integration, and Iterative Cycles – function as the core characteristics of DT-PBL. These categories represent the psychological and processual structures that sustain the model. Self-confidence acts as a motivational catalyst, interactive reflection serves as a social-cognitive bridge, knowledge integration provides the operational core, and iterative cycles ensure continuous adaptation and validation. Together, they constitute a coherent pedagogical model that explains how DT-PBL drives learning processes and produces measurable outcomes.

The subsequent subsections examine each dimension using interview data and classroom evidence. The analysis traces their developmental trajectories across project phases and their impact on students' lived experiences, supported by illustrative quotes. This approach shows not only how DT-PBL learning unfolds dynamically but also how it translates into outcomes with educational and practical significance.

5.1. Self-confidence

This study identifies self-confidence as a characteristic of DT-PBL, structured across three interrelated dimensions: creative confidence, interpersonal confidence, and problem-solving confidence. These dimensions do not manifest uniformly across the cycle but emerge at different phases, indicating that confidence evolves dynamically rather than existing as a fixed trait. This trajectory illustrates how confidence functions as both a driver of learning and a measurable outcome of the DT-PBL process.

Creative confidence was evident in the early and middle phases, when DT practices such as user observation and empathy mapping required students to move beyond conventional thinking and engage with real-world problems. Students reported that these practices enhanced their curiosity, sharpened their attention to overlooked details, and encouraged them to propose untested solutions. For example, one participant noted that:

'The creative confidence training module helped me quickly integrate into the project ... particularly when understanding user needs. We discovered many details that we had previously overlooked. This also gave me a significant sense of achievement at the project's outset and increased my courage to face uncertainties and challenges in future projects.' (S3)

Such evidence suggests that creative confidence operates as an intrinsic motivator, fostering engagement in uncertain contexts and laying the foundation for subsequent problem-solving.

Interpersonal confidence emerged primarily during the exploration and problem-definition phases, when students confronted diverse perspectives and conflicting interpretations. Structured activities such as facilitated brainstorming and guided group dialogue gradually built trust and cohesion. As one participant explained:

'In the early stages of the project, team members were not well acquainted with one another. The teacher facilitated brainstorming sessions and focus groups to help us get to know each other. This situation helped establish a strong trust, particularly in the first three months.' (S9)

Therefore, this study interprets interpersonal confidence as a relational resource, socially constructed through negotiation and trust-building, enabling students to transform initial tensions into productive collaboration.

Problem-solving confidence emerged during the implementation phase, as students transitioned from ideation to real-world application. Unlike the earlier forms, this confidence was grounded in evidence from rapid prototyping, iterative testing, and practical validation. As one participant explained, 'Through continuous, low-cost trial and error, we found a cost-effective alternative with similar performance' (S1). This indicates that problem-solving confidence was based on experiential mastery, shaped by cycles of failure and refinement that ultimately led to success.

Taken together, these findings show that self-confidence in DT-PBL develops through an iterative process of reinforcement and transformation. Creative confidence motivates engagement with uncertainty, interpersonal confidence sustains collaboration across diversity, and problem-solving confidence consolidates learning into validated solutions. This progression highlights self-confidence as both a structural driver and an outcome of DT-PBL, bridging motivation, process, and results, and underscoring its central role in enabling students to persist in the face of uncertainty, negotiate collaboration, and deliver context-sensitive innovations.

5.2. Interactive reflection

This study identifies interactive reflection as a central process in DT-PBL, enacted through team, teacher-student, and social interactions. Across all phases of the cycle, students engaged in reflective dialogue that connected experiential learning with both theoretical knowledge and real-world application.

Team interactions were most evident during the exploration and thinking phases. Students reported that collaborative discussions enabled them to reconcile divergent interpretations and develop more robust, collectively owned solutions. For example, one participant described how heated debates over technical options ultimately produced an integrated outcome that balanced multiple perspectives.

'During a discussion on a technical issue, there was a disagreement among team members, each proposing different solutions, leading to a heated discussion. However, through continuous communication and debate, we reflected on each other's suggestions, read further, and expanded our knowledge base. Ultimately, we integrated everyone's viewpoints to find a solution that balanced all opinions.' (S5)

These findings suggest that reflection was not an isolated act of retrospection but a socially embedded process in which disagreement, when managed constructively, became a catalyst for shared understanding and improved solutions. In this sense, interactive reflection functioned as a form of peer-to-peer learning, where individual insights were sharpened and validated through collective negotiation.

Teacher–student interactions served as an essential scaffold when teams encountered conceptual or technical obstacles. Rather than providing direct answers, instructors used guided questioning to help students articulate their challenges and explore alternative approaches. One participant noted that repeated cycles of such guidance shifted their stance from passively awaiting solutions to actively pursuing them through inquiry.

‘When we encountered a technical bottleneck while designing a new product, our mentor did not directly provide us with a solution but guided us through a series of questions. Through multiple interactions and reflections with the mentor, I began to understand the deeper meaning behind his questions and learned how to seek answers proactively. This guiding approach helped us find a solution and facilitated our deep thinking and reflection.’ (S7)

This highlights interactive reflection as a training ground for metacognitive growth. Students learned to step outside of immediate problems, monitor their reasoning, and take ownership of the inquiry process.

Social interactions with community members and stakeholders further extended reflective practice beyond the classroom. Feedback from real-world users prompted students to re-examine assumptions, refine design parameters, and prioritise practical feasibility. In one co-creation workshop, students adapted their prototype after discovering conflicts with local usage patterns (S8). These interactions introduced authenticity and unpredictability, compelling students to acknowledge the limits of classroom reasoning and cultivate socio-technical sensitivity and adaptability.

Taken together, the analyses show that interactive reflection in DT-PBL operates as a dynamic, multi-level feedback system. At the team level, it mediates differences and generates shared mental models. At the teacher–student level, it scaffolds metacognition and supports productive struggle. At the stakeholder level, it grounds learning in authentic contexts. This layered structure highlights reflection as an integrative bridge that links knowledge acquisition with application, ensuring learning remains adaptive and responsive to evolving challenges.

5.3. Knowledge integration

This study identifies knowledge integration as the central cognitive mechanism of DT-PBL, linking reflection with iteration through a recursive cycle of information exchange, knowledge transformation, and practical testing. Rather than a static transfer, integration unfolded as a dynamic process in which information was reorganised into new understandings and validated through authentic application. In this way, knowledge integration functioned simultaneously as the cognitive engine of learning and as a visible marker of how theory was enacted in practice.

Information exchange was most prominent during the exploration and thinking stages. Students collected and shared data through peer dialogue, stakeholder interviews, and literature reviews, gradually developing a comprehensive view of the problem space. As one participant observed, ‘The first step in the exploration process was to clarify the problem ... repeated communication and analysis revealed that our initial assumptions were inaccurate’ (S10). These exchanges did more than collect data; they connected new information with existing knowledge, forming frameworks that reduced redundancy and created opportunities for innovation.

Knowledge transformation characterised the transition from abstract analysis to systematic design. Students reported reorganising theoretical concepts in light of contextual demands and combining disciplinary insights with practical needs to create workable strategies. ‘We were able to combine theoretical knowledge with the need to solve problems in real contexts to form systematic solutions’ (S12). Continuous prototyping and redesign further refined these ideas, turning abstract knowledge into strategies suitable for real-world application. Thus, integration operated as a process of internalisation, reconfiguring theoretical insights into actionable know-how.

Practical testing dominated the practice and confirmation stages. Prototypes were implemented in authentic contexts, enabling students to assess feasibility, expose hidden flaws, and incorporate

iterative improvements. As one participant explained, ‘We proposed several solutions, then tested their feasibility through real-world application, which guided us to the most viable option’ (S14). This process validated existing knowledge while generating new insights, pushing the cycle back for further rounds of information exchange and transformation. Trial and error functioned as a dynamic filter, cultivating adaptability and strengthening confidence in solutions.

Taken together, these processes created a continuous knowledge cycle in which testing informed new problem definitions and transformation shaped subsequent iterations. Learning was not linear but evolved through constant adjustment, gaining significance only when enacted, critiqued, and reapplied. Notably, integration was accelerated through team collaboration, in which diverse disciplinary perspectives were combined to produce more prosperous and resilient solutions. In this sense, knowledge integration in DT-PBL was not merely the exchange of information but the construction of a living system of knowledge practices that enabled students to move fluidly between divergent exploration and convergent application. This mechanism supported both technical competence and transferable skills, such as collaboration, adaptability, and reflective judgment, which are essential for sustainable problem-solving in engineering contexts.

5.4. Iterative cycles

This study identifies iteration as a defining characteristic of DT-PBL and a key mechanism for sustaining deep learning. Unlike linear models, iteration functioned as a spiral, consolidating prior understanding while opening new problem spaces for exploration. Through this recursive dynamic, students not only refined project outcomes but also cultivated adaptive expertise, resilience, and long-term problem-solving capacity.

The analysis revealed three interrelated phases of iteration – problem identification, trial and error, and meaning update – each aligning with the goals of different stages of the DT-PBL process. In the early phase, students engaged in social exploration and rapid prototyping to clarify design directions and uncover user needs. As one participant recalled, ‘The first step in the exploration process was to clarify the problem ... repeated communication and analysis revealed that our initial assumptions were inaccurate’ (S10). This stage demonstrated that problem identification was not limited to defining tasks but involved reframing uncertainty into solvable challenges, thereby fostering empathy and flexibility.

Trial and error dominated the middle phase, where students tested ideas, received feedback, and refined their solutions in continuous cycles. One student reflected, ‘My most significant takeaway has been learning to grow through trial and error ... each error provided feedback that clarified the underlying issues. Through continuous adjustments, we ultimately arrived at a better solution’ (S11). In this process, failure was redefined as formative feedback, transforming setbacks into drivers of progress. This finding aligns with Hatano and Inagaki’s (2013) perspective, which posits that adaptive expertise develops through repeated encounters with novelty and difficulty. Iteration thus recalibrated both students’ technical strategies and their confidence, reinforcing the idea that mastery in engineering requires embracing, rather than avoiding, uncertainty.

In the later stages, students engaged in meaning updates, reinterpreting prior experiences to align with evolving project demands. One participant emphasised, ‘Establishing a robust process through stage-specific reflection and adjustments allows us to better adapt to complex real-world situations and design more sustainable and long-term solutions’ (S13). Meaning updates ensured that outcomes were not static solutions but evolving frameworks for future innovation, consistent with Nichols et al. (2013), who highlight the importance of systems-level feedback loops for sustainable innovation.

Taken together, these phases illustrate how iteration operated as the connective tissue of DT-PBL. It linked self-confidence with mastery experiences, embedded reflection into action, and validated knowledge integration through practice. By embedding iterative cycles across phases, DT-PBL sustained learning as a recursive and adaptive process.

5.5. Integrated analysis

The integrated analysis of the four categories reveals a coherent DT-PBL model in which motivational, reflective, cognitive, and adaptive mechanisms interact dynamically to shape students' learning trajectories and outcomes. Self-confidence initiates engagement by motivating learners to enter uncertain problem spaces; interactive reflection translates this motivation into adaptive inquiry; knowledge integration structures information into actionable solutions; and iterative cycles sustain and recalibrate these processes through feedback and meaning-making. Collectively, these findings suggest that the effectiveness of DT-PBL lies not in any single element, but in the interdependence of these mechanisms, which together constitute a cyclical, self-reinforcing learning ecology.

6. Discussion

This study aimed to identify the characteristics and mechanisms of DT-PBL and to examine how these multidimensional and mechanisms shape students' learning processes (RQ2) and outcomes (RQ3). Through grounded theory analysis, four interrelated dimensions were distilled: self-confidence, interactive reflection, knowledge integration, and iterative cycles. These dimensions transcend surface-level classroom activities, capturing the psychological and procedural mechanisms that underpin DT-PBL as a closed-loop learning ecology. This conceptualisation provides the empirical foundation for a coherent pedagogical model that addresses RQ1.

Regarding the dynamics of learning (RQ2), the findings reveal that learning in DT-PBL evolves recursively rather than linearly, diverging from traditional PBL models that follow a fixed 'planning–execution–presentation' sequence (Al-Busaidi, Yusuf, and Reinders 2021). Each dimension becomes salient at different stages, while remaining mutually interdependent, forming a cyclical and self-reinforcing system. Self-confidence sustains engagement and persistence; interactive reflection and knowledge integration function as relational and cognitive mechanisms that facilitate meaning-making and complexity management; and iterative cycles provide continuous feedback that maintains learning momentum. Collectively, these mechanisms demonstrate how DT-PBL fosters a dynamic learning ecology that cultivates creativity, reflective judgment, and adaptive problem-solving – key competencies for engineering innovation.

Within this recursive structure, self-confidence emerged as the motivational engine of learning. It initially appeared as creative confidence during early exploration, nurtured through empathy-driven inquiry and user-centered engagement – consistent with Rohmaniyah and Asih (2024), who showed that creative capacity can be intentionally developed in open-ended design contexts. As projects progressed, confidence diversified across interpersonal and problem-solving domains. Mid-phase collaboration fostered interpersonal confidence, aligning with Rachmad (2025), who demonstrated that interdisciplinary teamwork cultivates trust and collective creative ownership. In later phases, students exhibited greater problem-solving confidence as they applied cross-disciplinary knowledge in real-world prototyping, echoing Snape and Fox-Turnbull's (2013) findings that authentic implementation reinforces both technical mastery and design agency.

Unlike earlier research viewing confidence as a fixed personal trait (Burns, Burns, and Ward 2016), our findings characterise it as a dynamic and developmental state, evolving from creative to interpersonal to problem-solving confidence through iterative engagement. This perspective extends Bandura's (1997) self-efficacy framework by embedding confidence in a socially constructed, experiential context. Pedagogically, confidence should thus be regarded not as a prerequisite for performance but as a context-dependent outcome of DT-PBL learning, strengthened through structured reflection and formative feedback.

Interactive reflection functioned as the bridge between motivation and sustained learning. Team dialogues and debates enabled students to challenge assumptions and co-construct shared mental models, thereby enhancing coordination (Nobutoshi 2023). Instructors reinforced this process

through probing questions rather than direct instruction, transforming students from passive recipients into active inquirers – a practice consistent with Thomas et al. (2025), who showed that exploratory questioning deepens engagement and critical thinking. Moreover, stakeholder participation anchored reflection in authentic contexts, compelling students to reconcile theoretical understanding with practical constraints. This interaction cultivated socio-technical awareness and empathy, echoing Avsec et al. (2024), who found that stakeholder dialogue enhances contextual sensitivity and human-centered reasoning in engineering design.

Departing from studies that conceptualised reflection as an individual cognitive process (Jordi 2011), this study redefines it as a multi-layered feedback system – cognitive, social, and contextual – that continuously anchors and propels learning. This interpretation extends Kolb et al.'s (2014) experiential model by embedding reflection within collective negotiation and co-creation, revealing that metacognition in engineering education is socially constructed rather than introspective. Consequently, educators should design structured social reflection loops that integrate technical reasoning with human-centered interpretation, transforming reflection into a generative mechanism for collaborative innovation.

Knowledge integration emerged as the cognitive engine of DT-PBL. Learning unfolded not as static information transfer but as a recursive cycle of exchange, transformation, and validation. Peer collaboration and stakeholder dialogue broadened problem framing by incorporating multiple disciplinary perspectives, consistent with Feng (2025), who demonstrated that interdisciplinary negotiation drives conceptual change. Knowledge transformation occurred as students reinterpreted theoretical insights within contextual constraints, reorganising them into actionable design strategies. Prototyping, then externalising and testing these ideas, was refined through iterative feedback – paralleling Deininger et al. (2019), who identified prototyping as a key mechanism for integrating tacit and explicit engineering knowledge.

Unlike prior research that viewed integration as mere cognitive synthesis (Palermos 2014), our findings depict it as a socio-cognitive ecosystem – a system of information exchange, collaborative negotiation, and reflective testing forming continuous loops of internalisation. This reconceptualization extends Nonaka, Takeuchi, and Umemoto's (1996) SECI model by situating knowledge creation within socially mediated design activity. Theoretically, it offers a systemic account of how project-based design learning transforms theory into situated practice; practically, it suggests that curricula should create cross-boundary integration spaces where interdisciplinary teams co-develop, validate, and refine design logic through structured feedback and iterative testing.

Iterative cycles sustained and amplified all prior mechanisms, functioning as the connective tissue maintaining systemic momentum. Students repeatedly engaged in problem identification, experimentation, and meaning reconstruction, recalibrating confidence, reflection, and integration at each turn. This process resonates with Hatano and Inagaki's (2013) concept of adaptive expertise, where flexibility and resilience emerge through iterative encounters with novelty and failure. Unlike traditional PBL models that treat iteration as procedural repetition (Vogeler et al. 2022), this study positions DT-PBL iteration as a developmental mechanism. This feedback-driven sense-making process transforms setbacks into learning opportunities and fosters sustainable growth of understanding (Nichols et al. 2013). Iteration thus becomes not a discrete stage but a pervasive mindset of reflective experimentation embedded across all phases of learning.

Together, these intertwined mechanisms address RQ2: learning in DT-PBL unfolds as a recursive, dynamic process in which confidence fuels engagement, reflection mediates meaning-making, integration structures knowledge into actionable insight, and iteration sustains adaptation and renewal.

Regarding learning outcomes (RQ3), DT-PBL fosters outcomes beyond the production of functional prototypes. Student interviews revealed substantial growth in transferable capabilities – including reflective judgment, socio-technical sensitivity, collaborative resilience, and sustainable problem-solving. In other words, students achieved not only immediate project results but also enduring cognitive and affective dispositions that shape how they approach uncertainty and complexity. These outcomes emerge from the interaction of the four core dimensions – confidence,

reflection, integration, and iteration – that form a self-reinforcing learning ecology. Confidence enabled persistence amid uncertainty; reflection strengthened metacognitive and dialogic capacities; knowledge integration translated theory into contextualised practice; and iteration embedded adaptability and resilience. Unlike prior studies emphasising short-term project performance (Biedenbach and Müller 2012), this study underscores DT-PBL's role in cultivating long-term cognitive and affective transformations – changes in how students think, collaborate, and act in the face of uncertainty. Accordingly, DT-PBL should be designed not merely to generate innovative products but to develop innovation-ready engineers capable of navigating complex, ill-structured challenges through reflection-driven, integrative, and adaptive thinking.

In sum, this discussion demonstrates how the four core dimensions of DT-PBL collectively address all three research questions. They constitute the structural mechanisms of a coherent pedagogical model (RQ1), explain how learning unfolds recursively across project phases (RQ2), and reveal the transferable, measurable outcomes that mark genuine educational change (RQ3). Conceptualising DT-PBL as a cyclical, self-reinforcing learning ecology highlights its distinctive value in engineering education: it does not simply train students to complete predefined tasks, but equips them to face the complexities of real-world innovation with confidence, reflective capacity, and adaptive problem-solving skills.

7. Conclusion

The DT-PBL framework provides engineering educators with a structured pathway to integrate human-centered creativity with rigorous technical problem-solving. The four-phase structure – exploration, thinking, practice, and confirmation – provides a replicable template for course design that bridges theoretical knowledge with authentic engineering challenges. The integration of design thinking strategies enables educators to scaffold both divergent and convergent thinking, ensuring that innovation is balanced with feasibility and technical precision.

From a curriculum design perspective, the findings suggest that the DT-PBL framework in engineering projects enhances not only technical competencies but also communication, adaptability, and leadership skills. These transferable skills are increasingly critical for engineering graduates operating in complex, interdisciplinary environments. Furthermore, the emphasis on iterative cycles and meaning updates aligns well with engineering's need for continuous improvement and lifecycle thinking, making DT-PBL particularly suited to sustainable innovation education.

7.1. Limitations and future research directions

While this study offers valuable insights into the mechanisms of DT-PBL, several limitations should be acknowledged. First, the participant sample was drawn from a single institutional and cultural context, which may limit the generalizability of findings across diverse engineering programmes or international settings. Future studies should therefore examine DT-PBL implementations across different institutional types and disciplinary domains to capture contextual variations. Second, although the qualitative design effectively revealed processual depth, it constrained the ability to trace developmental trajectories or infer causal relationships over time. Integrating longitudinal or quasi-experimental designs could help assess the sustained effects of DT-PBL on students' competence development, creativity, and innovation capacity. Third, reliance on self-reported, retrospective interview data introduces potential response bias, particularly regarding students' perceived learning outcomes. Triangulating these accounts with classroom observations, performance analytics, or artifact-based assessments would enhance methodological rigour and validity. Fourth, from a methodological standpoint, this study prioritised interpretive depth over analytical generalisation. Future research should adopt multi-method approaches – such as combining grounded theory with learning analytics or behavioural sequence analysis – to reveal how DT-PBL functions as an evolving learning system rather than a static instructional framework. Finally, the

increasing integration of digital collaboration tools, AI-assisted design environments, and virtual prototyping platforms presents new opportunities to extend DT-PBL beyond physical classrooms. Investigating how these technologies mediate reflection, iteration, and collaboration can yield valuable insights for scaling and adapting DT-PBL within global and hybrid engineering education contexts.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The Anhui Provincial Department of Education supported this work under Grant (2022jyxggyj284) and Anhui Provincial Higher Education Research Plan Project (2022AH050544) and Anhui University of Finance and Economics courses (ackcszk2024015).

Ethical approval

This work involved human subjects or animals in its research. The Anhui University of Finance and Economics School of Arts of Review Board approved all ethical and experimental procedures and protocols.

Informed consent

There was no direct contact with participants in this study, and thus, informed consent was not required.

Data availability statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

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